

Equating Two Sequential BinoMOS Stages with a Single BinoMOS Stage

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We consider two BinoMOS operations in sequence. The output of the first stage is used as the true quality input to the second stage. This models a situation encountered in our work on learning through the noise. There, MOS values from datasets were treated as true quality values and used as input to BinoMOS in order to add more vote noise. But the original MOS values are based on a finite number of votes, so instead of treating them as true quality, it is more appropriate to model them as BinoMOS. This leads to notion of two sequential BinoMOS stages and we need to know the effective number of votes per file at the output of the second stage. This will be a function of the number of votes per file used in each of the two stages. The problem at hand is presented graphically in Fig. 1.

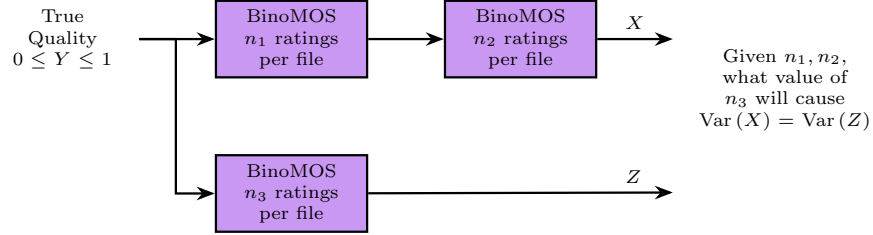


Figure 1: Problem definition.

In order to keep the math as clean and simple as possible, we use $0 \leq Y \leq 1$ without loss of generality. In the interest of generality, we allow vote values in $\{0, 1, \dots, n_d\}$ and thus MOS values are in $[0, n_d]$.¹ The work that follows is converted to the common scale by setting $n_d = 4$ and adding a constant value of 1.0 to every vote and every MOS.²

First, we find the variance of the random variable Z in Fig. 1 which is produced by a single BinoMOS process. The vote distribution is $\text{Binomial}(n_d, Y)$. The distribution of the sum of n_3 of these votes is $\text{Binomial}(n_d n_3, Y)$ and the variance of this distribution is $n_d n_3 Y(1 - Y)$. This sum must be scaled by $\frac{1}{n_3}$ to arrive at the BinoMOS random variable Z which has a variance of

$$\text{Var}(Z) = \frac{n_d n_3 Y(1 - Y)}{n_3^2} = \frac{n_d Y(1 - Y)}{n_3}. \quad (1)$$

Next we find the variance the random variable X which is produced by two BinoVote processes in sequence as shown in Fig. 1. The derivation is aided by expanding the upper path in Fig. 1 to show the Binomial processes that underlie the BinoMOS processes and the associated summing and scaling. This expansion is given in Fig. 2.

¹The symbol n_d was selected because it gives the number of *deltas*, or steps between the votes. In prior work we defined n_s to be the number of allowed vote values. This resulted in the expression $n_s - 1$ appearing a lot. Here we use $n_d = n_s - 1$ for simplicity.

²The addition of a constant does not change any variance calculation, but it does clutter the math considerably.

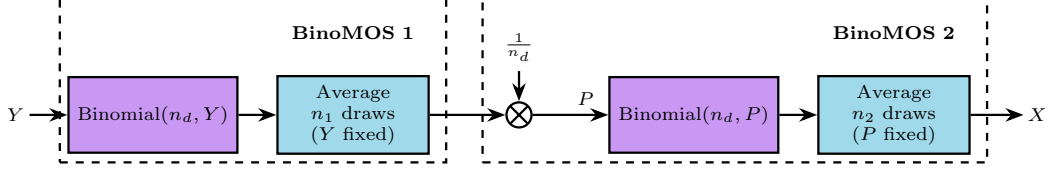


Figure 2: Full details of two BinoMOS stages that comprise the upper path of Fig. 1.

The mean of $\text{Binomial}(n, p)$ random variable is np . Thus it is easy to show that $\mathbb{E}(P|Y) = Y$ and that $\mathbb{E}(X|P) = n_d P$. Similar to (1), it is easy to show that

$$\text{Var}(P) = \frac{n_d n_1 Y(1-Y)}{n_d^2 n_1^2} = \frac{Y(1-Y)}{n_d n_1}. \quad (2)$$

The law of total variance gives

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}(\text{Var}(X|P)) + \text{Var}(\mathbb{E}(X|P)) \\ &= \mathbb{E}\left(\frac{n_d n_2 P(1-P)}{n_2^2}\right) + \text{Var}(n_d P) \\ &= \frac{n_d}{n_2} \mathbb{E}(P(1-P)) + n_d^2 \text{Var}(P) \\ &= n_d \left(\frac{1}{n_2} \mathbb{E}(P) - \frac{1}{n_2} \mathbb{E}(P^2) + n_d \text{Var}(P) \right) \\ &= n_d \left(\frac{1}{n_2} \mathbb{E}(P) - \frac{1}{n_2} (\text{Var}(P) + \mathbb{E}(P)^2) + n_d \text{Var}(P) \right) \\ &= n_d \left(\frac{1}{n_2} (\mathbb{E}(P) - \mathbb{E}(P)^2) + \left(n_d - \frac{1}{n_2}\right) \text{Var}(P) \right) \\ &= n_d \left(\frac{1}{n_2} (Y - Y^2) + \left(n_d - \frac{1}{n_2}\right) \text{Var}(P) \right). \end{aligned} \quad (3)$$

Using (2) in (3) gives

$$\begin{aligned} \text{Var}(X) &= n_d \left(\frac{Y(1-Y)}{n_2} + \left(n_d - \frac{1}{n_2}\right) \left(\frac{Y(1-Y)}{n_d n_1} \right) \right) \\ &= n_d Y(1-Y) \left(\frac{1}{n_2} + \left(\frac{n_d - \frac{1}{n_2}}{n_d n_1} \right) \right) \\ &= n_d Y(1-Y) \left(\frac{1}{n_1} + \frac{1}{n_2} - \frac{1}{n_1 n_2 n_d} \right). \end{aligned} \quad (4)$$

Given n_1 and n_2 , we wish to select n_3 so that $\text{Var}(Z) = \text{Var}(X)$. When we compare (1) with (4) it is easy to see that the solution is

$$\frac{1}{n_3} = \left(\frac{1}{n_1} + \frac{1}{n_2} - \frac{1}{n_1 n_2 n_d} \right), \quad (5)$$

or equivalently

$$n_3 = \frac{1}{\left(\frac{1}{n_1} + \frac{1}{n_2} - \frac{1}{n_1 n_2 n_d} \right)}. \quad (6)$$

For example, let $n_d = 4$ for the case of the common 1 to 5 scale. Use real MOS values with $n_1 = 24$ votes per file as true quality values in BinoMOS with $n_2 = 10$ votes per file. If we assume that BinoMOS is a

good model for the real MOS values, then the resulting vote noise is equivalent to that of a single BinoMOS process with

$$n_3 = \frac{1}{\left(\frac{1}{24} + \frac{1}{10} - \frac{1}{24 \cdot 10 \cdot 4}\right)} \approx 7.11 \quad (7)$$

votes per file.